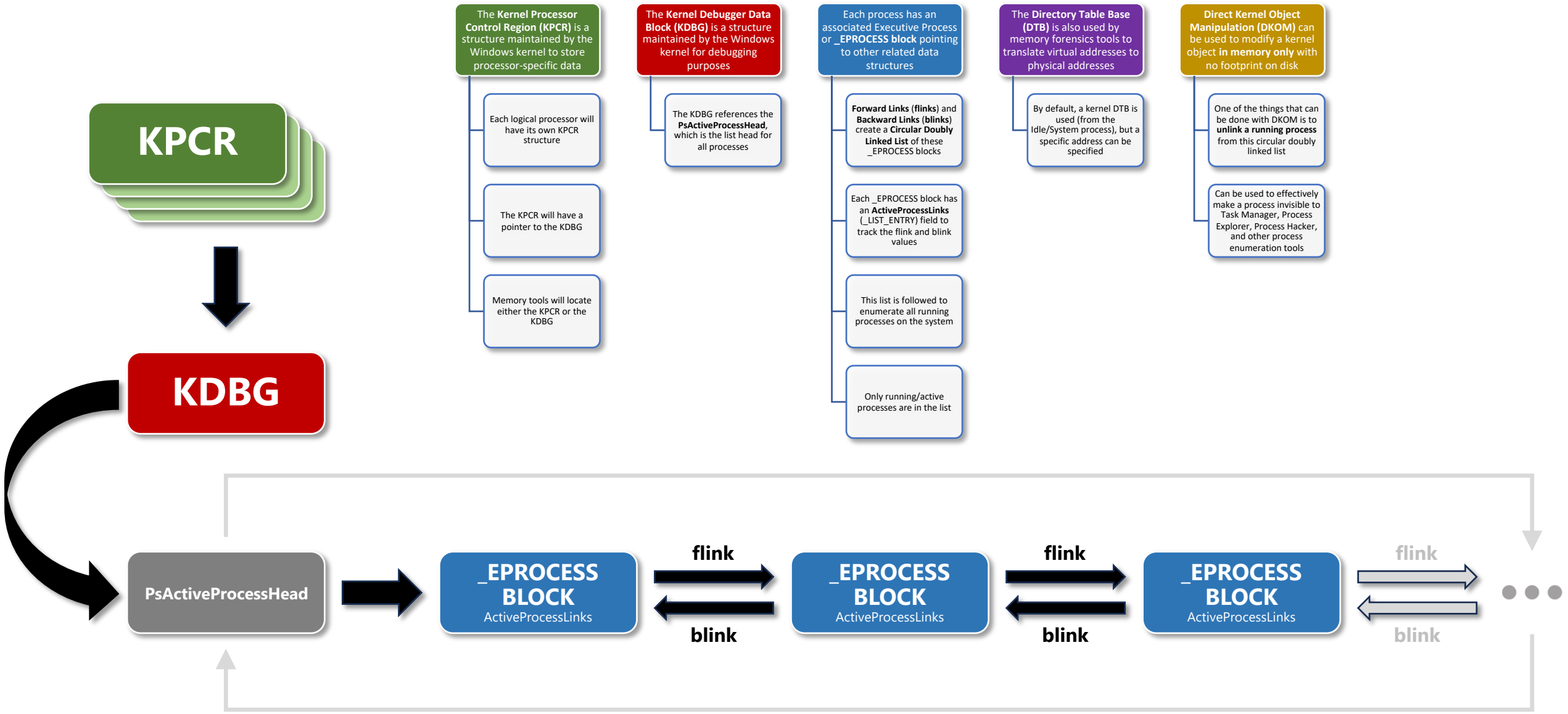
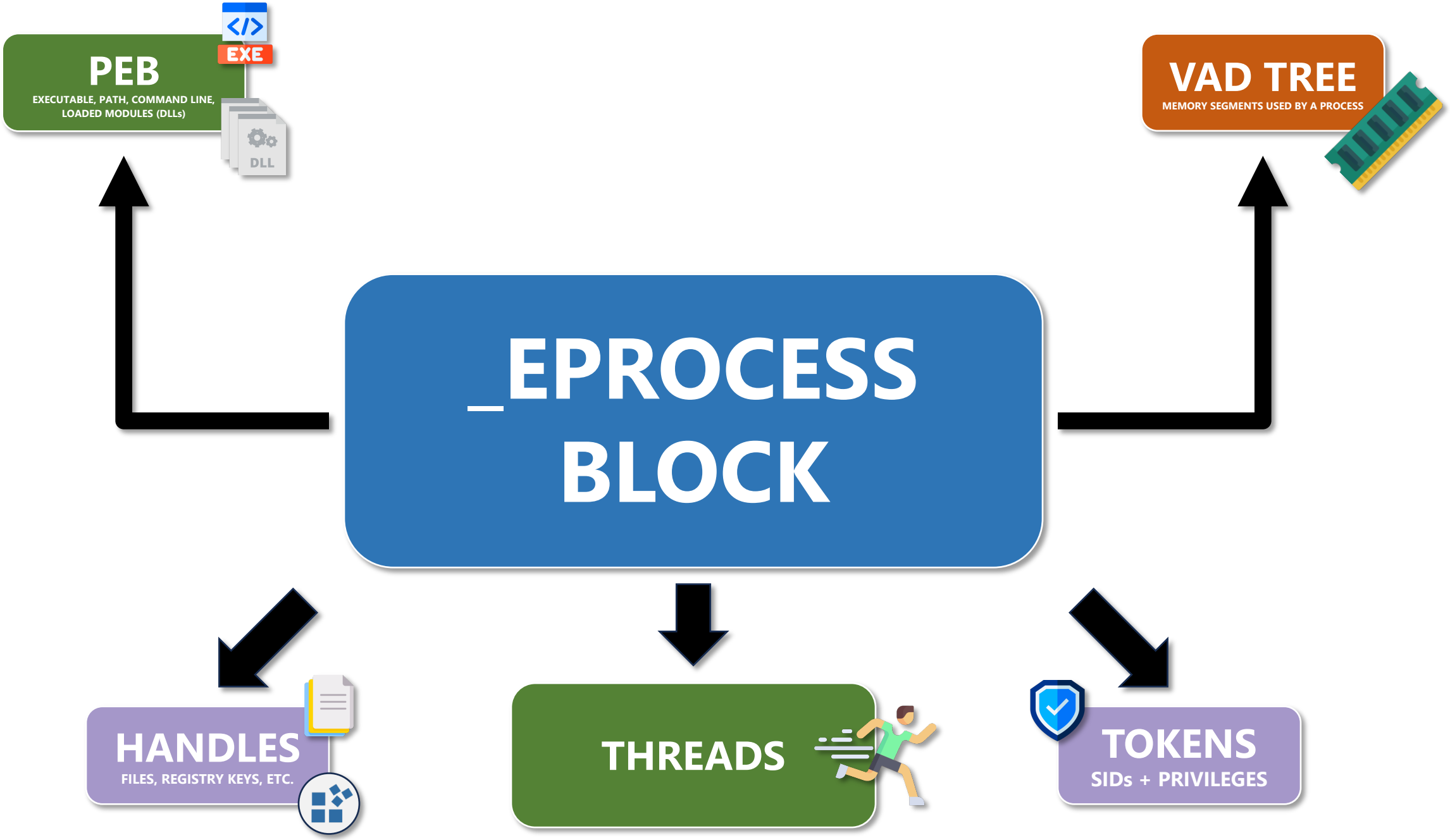
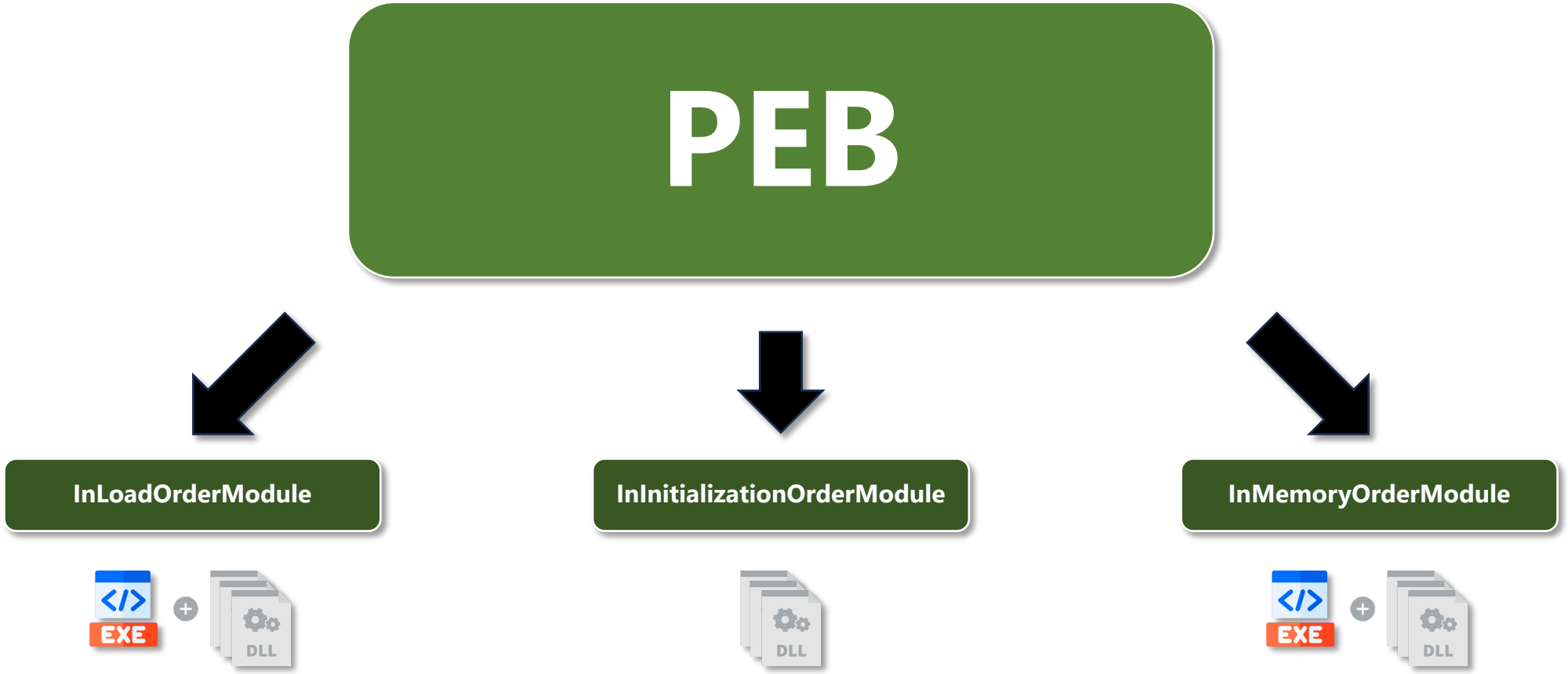
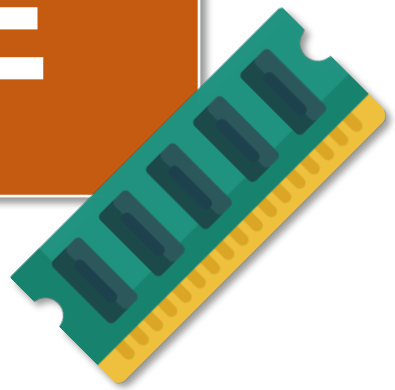

KPCR
KDBG
PsActiveProcessHead
The Kernel Processor Control Region (KPCR) is a structure maintained by the Windows kernel to store processor-specific data
Each logical processor will have its own KPCR structure
The KPCR will have a pointer to the KDBG
Memory tools will locate either the KPCR or the KDBG
The Kernel Debugger Data Block (KDBG) is a structure maintained by the Windows kernel for debugging purposes
The KDBG references the PsActiveProcessHead, which is the list head for all processes
Each process has an associated Executive Process or _EPROCESS block pointing to other related data structures
Forward Links (flinks) and Backward Links (blinks) create a Circular Doubly Linked List of these _EPROCESS blocks
Each _EPROCESS block has an ActiveProcessLinks (_LIST_ENTRY) field to track the flink and blink values
This list is followed to enumerate all running processes on the system
Only running/active processes are in the list
The Directory Table Base (DTB) is also used by memory forensics tools to translate virtual addresses to physical addresses
By default, a kernel DTB is used (from the Idle/System process), but a specific address can be specified
Direct Kernel Object Manipulation (DKOM) can be used to modify a kernel object in memory only with no footprint on disk
One of the things that can be done with DKOM is to unlink a running process from this circular doubly linked list
Can be used to effectively make a process invisible to Task Manager, Process Explorer, Process Hacker, and other process enumeration tools
_EPROCESS BLOCK
ActiveProcessLinks
flink
blink
_EPROCESS BLOCK
ActiveProcessLinks
flink
blink
_EPROCESS BLOCK
ActiveProcessLinks
flink
blink


PEB
EXECUTABLE, PATH, COMMAND LINE, LOADED MODULES (DLLs)
EXE
DLL
VAD TREE
MEMORY SEGMENTS USED BY A PROCESS
_EPROCESS BLOCK
HANDLES
FILES, REGISTRY KEYS, ETC.
THREADS
TOKENS
SIDs + PRIVILEGES


PEB
InLoadOrderModule
InInitializationOrderModule
InMemoryOrderModule
EXE
DLL
DLL
EXE
DLL

VAD TREE



MappedPath